Calibrated prediction intervals (Tier 1, items 20, 21, 24, 29)

One-sided **upper** bounds on `scaling_factor_p50`, evaluated leave-one-machine-out. Coverage is `P(y <= upper_bound)`; it should be at least the nominal level. `rel width` is `upper_bound / actual`, averaged — lower is tighter, 1.0 would be a perfect oracle bound.

method	nominal	empirical (mean over 5 folds)	worst single machine	rel width
gbr_log	90%	66.0% ⚠	56.0%	1.180
gbr_log	95%	72.9% ⚠	61.9%	1.243
gbr_log	99%	85.2% ⚠	74.4%	1.402
gbr_quantile	90%	78.1% ⚠	70.2%	1.365
gbr_quantile	95%	84.6% ⚠	79.8%	1.457
gbr_quantile	99%	91.5% ⚠	79.8%	1.865
split_conformal	90%	91.2%	78.0%	1.625
split_conformal	95%	97.0%	90.5%	1.871
split_conformal	99%	99.6%	98.8%	2.331
weighted_conf	90%	91.8%	78.0%	1.619
weighted_conf	95%	97.0%	90.5%	1.890
weighted_conf	99%	99.8%	98.8%	2.380

Reading this

⚠ marks a method that fails to deliver its promised coverage.

The measurement-noise floor on the target itself is **0.60%** relative standard error (item 29): `y` is a ratio of two p50s each estimated from 19 warm repetitions, so even a perfect model cannot resolve `y` more finely than this. Any interval tighter than that is claiming precision the measurements do not contain.

`split_conformal` calibrates on a training machine the regressor never saw, so its guarantee is conditional on the target machine being exchangeable with that calibration machine. The c5a result is direct evidence that assumption fails for an outlier machine — which is exactly what `weighted_conf` corrects for, by weighting calibration points by hardware-signature proximity to the target. Compare their worst-single-machine columns rather than their means: the mean hides the fold that matters.